This is not a big step from the Newtonian form of the equations of motion $\dot{\boldsymbol{q}} = \boldsymbol{v}; \boldsymbol{M}\dot{\boldsymbol{v}} = \boldsymbol{F}$. More generally, for molecular models, the Hamiltonian and Lagrangian formulations are interchangeable, but the use of the Hamiltonian form is preferred for allowing simplified description of the geometric character of the solutions of the system as we discuss in Chaps. 2–4.

The set of all positions and momenta for which the energy is finite is termed the *phase space*. The instantaneous state of a molecular system involving many, say N, particles moving in $\mathbb{R}^3$ is described by coordinates and positions, i.e., by a point in $\mathbb{R}^{6N}$. The "points" defining atomic positions (generally taken to be located in $\mathbb{R}^3$) should be clearly distinguished from the "points," in $\mathbb{R}^{6N}$, of the phase space, which represent the collection of all position and momenta vectors of the atoms. One may speak of a "phase space point" or simply "phase point" to describe the latter. We are ultimately going to be interested in how sets of phase points are carried forward by solution of the differential equations. It is crucial to observe that when we discuss such sets the individual points therein are understood to be completely independent of one another, and their motion is governed only by the laws of mechanics as described by the equations of motion.

1.5 Existence and Uniqueness of Solutions

One of the most important properties of a typical classical molecular Hamiltonian system is the existence and uniqueness of solutions started from a generic initial condition. The traditional study of existence and uniqueness for systems of nonlinear ordinary differential equations gives local results only, but for Hamiltonian systems considered in molecular dynamics the solutions are typically globally defined. The uniqueness of solutions is easily verified in the usual way (as for the local result for uniqueness of solutions). The key point is that, with the energy constraint, solutions typically remain bounded for all time.

Assume that U is a potential energy function which is bounded below, $U \geq U_{\min}$. For given $\mathrm{E}_0 \geq U_{\min}$ define $\Sigma_{\mathrm{E}_0} = \{(\boldsymbol{q},\boldsymbol{p}) | H(\boldsymbol{q},\boldsymbol{p}) = \mathrm{E}_0\}$. Then, for $(\boldsymbol{q},\boldsymbol{p}) \in \Sigma_{\mathrm{E}_0}$ we have

$$\frac{\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p}}{2} + U(\boldsymbol{q}) = \mathrm{E}_0 \Rightarrow \frac{\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p}}{2} = \mathrm{E}_0 - U(\boldsymbol{q}) \leq \mathrm{E}_0 - U_{\min}.$$

$\boldsymbol{M}^{-1}$ is a positive definite matrix (assumed here to be constant), so we can infer that the momenta are bounded at fixed total energy. We would like to say something similar for positions. We have, at energy E_0,

$$U_{\min} \leq U(\boldsymbol{q}) \leq \mathrm{E}_0.$$

What is needed is an assumption that the level sets $\hat{\Sigma}_\alpha = \{\boldsymbol{q} | U(\boldsymbol{q}) = \alpha\}$ are bounded uniformly for $\alpha \in [U_{\min}, \mathrm{E}_0]$. Then it follows that solutions satisfying the energy constraint remain confined to a compact (closed and bounded) set.

The assumption on U is not satisfied by some simple potentials. For example consider $U(x, y) = x^2$ which is completely independent of y and so places no restriction on that variable for constant energy. The condition on U means that it has a *confining* property. To ensure bounded solutions in systems that do not have a confining potential, we should add other assumptions, for example the assumption that the position vectors of each atom are restricted by periodic boundary conditions to lie on a three-dimensional torus (see Section 1.6).

1.5.1 The Flow Map

Consider now the initial value problem

$$\dot{\boldsymbol{z}} = \boldsymbol{f}(\boldsymbol{z}), \qquad \boldsymbol{z}(0) = \boldsymbol{\zeta}, \tag{1.5}$$

in a m-dimensional space. If we assume that $\boldsymbol{f}$ corresponds to a molecular Hamiltonian system satisfying the assumptions of the existence and uniqueness result of the previous subsection, then we may define a mapping from a point in phase space to the point t units later along the time-evolution starting from the initial point. We refer to this map as the *flow map* and denote it by $\mathscr{F}_t$. $\mathscr{F}_t(\boldsymbol{\zeta}) = \boldsymbol{z}(t)$ solves the initial value problem (1.5). It may equally describe the evolution of a point with time, or the evolution of a set of points with time.

The classical systems treated here can be solved forward or backward in time. Observe that $\mathscr{F}_{-t}\mathscr{F}_t = \mathrm{Id}$ (the identity map), thus the flow map is invertible and, indeed, the family of flow maps defined for different values of t form an Abelian group under the operation of composition ($\mathscr{F}_t\mathscr{F}_s = \mathscr{F}_s\mathscr{F}_t = \mathscr{F}_{t+s}$). The flow map of a Hamiltonian system conserves its Hamiltonian, thus

$$H(\mathscr{F}_t(\boldsymbol{\zeta})) \equiv H(\boldsymbol{\zeta}).$$

(This equation just expresses the conservation of energy.)

The flow map provides an easy notation for describing solutions (even in cases where it is difficult or impossible to compute it exactly). When we speak of numerical methods for solving differential equations, we generally mean methods that approximate the flow map in a suitable sense.

Despite the existence of a well-defined flow map, we cannot solve most systems of differential equations analytically (even Hamiltonian ones) so we cannot usually write out a formula for the flow map. An exception is the harmonic oscillator,

$$\dot{q} = p, \qquad \dot{p} = -\Omega^2 q,$$